Section 16.5: Curl and Divergence

Vector Calculus

Curl: Measuring Rotation in 3D

In 16.4, the 2D curl $Q_x - P_y$ measured local rotation. In 3D, we need a **vector** to capture the axis and rate of rotation.

Curl of a Vector Field

$$\text{curl } \vec{F} = \nabla \times \vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ P & Q & R \end{vmatrix}$$

Expand by cofactors (just like a 3×3 determinant):

$$= \vec{i}\left(\frac{\partial R}{\partial y} - \frac{\partial Q}{\partial z}\right) - \vec{j}\left(\frac{\partial R}{\partial x} - \frac{\partial P}{\partial z}\right) + \vec{k}\left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right)$$

Notice: the $\vec{k}$-component is $Q_x - P_y$ — the 2D curl from Green's Theorem.

Curl is a vector: its **direction** is the axis of rotation (right-hand rule), its **magnitude** is how fast the field spins. Use the determinant to get the signs right.

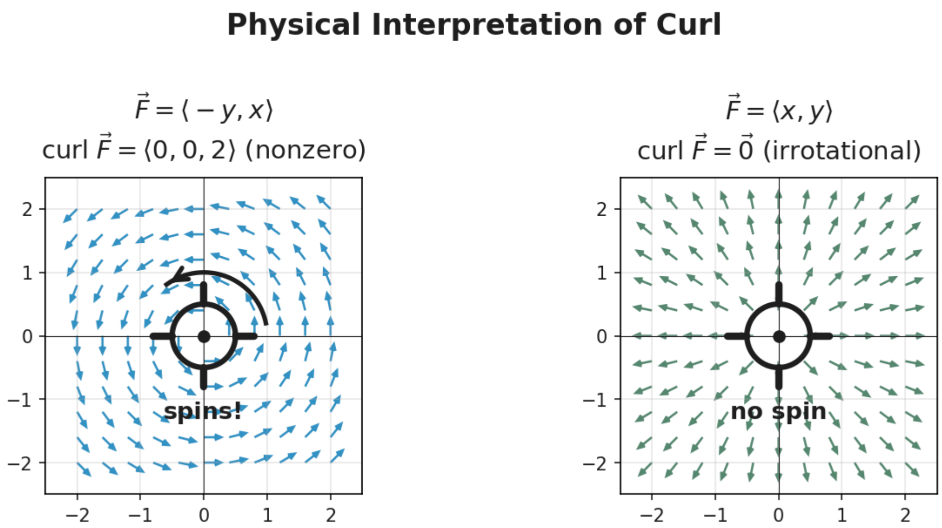
Example 1: Computing Curl

Example 1

Find $\nabla \times \vec{F}$ for $\vec{F}(x, y, z) = \langle x^2yz,\ xy^2z,\ xyz^2 \rangle$.

Set up the determinant with $P = x^2yz$, $Q = xy^2z$, $R = xyz^2$ and expand along the first row. Watch the sign on $\vec{j}$.

What Does Curl Mean Physically?



Paddle wheel test: Place a tiny paddle wheel in the flow.

- curl $\vec{F} \neq \vec{0}$: the paddle wheel **spins**. Direction of curl = axis of rotation.
- curl $\vec{F} = \vec{0}$: no spinning $\to$ **irrotational**

Example: $\vec{F} = \langle -y, x, 0 \rangle$ (rotation in xy-plane): $\nabla \times \vec{F} = \langle 0, 0, 2 \rangle$. Curl points in $+z$ (CCW from above), magnitude 2.

Example: $\vec{F} = \langle yz, xz, xy \rangle$: $\nabla \times \vec{F} = \vec{0}$. No rotation $\to$ conservative (potential: $f = xyz$).

Zero curl = no rotation = conservative (on simply connected domains). Nonzero curl = the field has local spinning.

Divergence: Measuring Expansion

Divergence of a Vector Field

$$\text{div}\, \vec{F} = \nabla \cdot \vec{F} = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z}$$

Much simpler than curl: differentiate each component with respect to its “own” variable and add.

Output types (remember dot vs cross):

- $\nabla \cdot \vec{F}$ (dot $\to$ scalar): divergence is a **number** at each point
- $\nabla \times \vec{F}$ (cross $\to$ vector): curl is a **vector** at each point

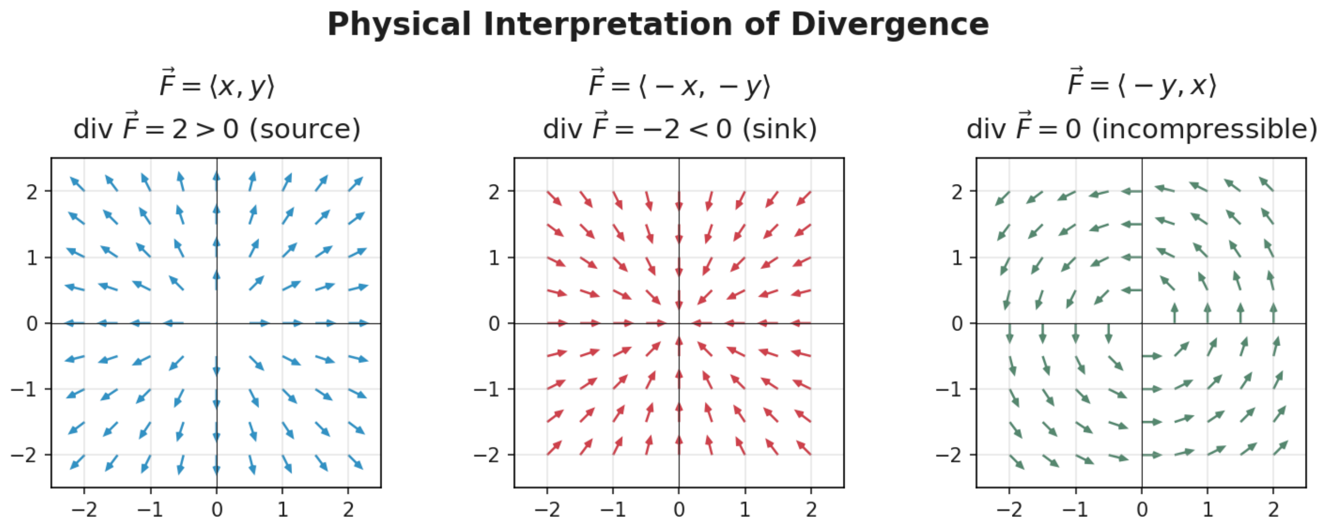
Example 2: Computing Divergence

Example 2

Find $\nabla \cdot \vec{F}$ for $\vec{F}(x, y, z) = \langle e^x \sin y,\ e^x \cos y,\ z^2 \rangle$.

Compute $P_x + Q_y + R_z$.

What Does Divergence Mean Physically?



Think of $\vec{F}$ as fluid flow:

- $\nabla \cdot \vec{F} > 0$: **source** — more flows out than in (expansion)
- $\nabla \cdot \vec{F} < 0$: **sink** — more flows in than out (compression)
- $\nabla \cdot \vec{F} = 0$: **incompressible** — what flows in equals what flows out

Example: $\vec{F} = \langle x, y, z \rangle$: $\nabla \cdot \vec{F} = 3$. Constant positive divergence everywhere — uniform expansion (like an explosion).

Divergence = “flux density.” Positive = source, negative = sink, zero = incompressible.

Conservative Test in 3D and Two Key Identities

Theorem: Curl Test for Conservative Fields

On an open, simply connected region: $\vec{F}$ is conservative $\iff \nabla \times \vec{F} = \vec{0}$.

This generalizes $P_y = Q_x$ to 3D — curl $= \vec{0}$ means **all three** cross-partial pairs match.

Two identities (both from Clairaut's Theorem — mixed partials cancel in pairs):

$$\nabla \times (\nabla f) = \vec{0} \qquad \text{and} \qquad \nabla \cdot (\nabla \times \vec{F}) = 0$$

- “Curl of a gradient is always zero” — every conservative field is irrotational
- “Div of a curl is always zero” — useful test: if $\nabla \cdot \vec{G} \neq 0$, then $\vec{G}$ cannot be a curl

The sequence: scalar $\xrightarrow{\nabla}$ vector $\xrightarrow{\nabla\times}$ vector $\xrightarrow{\nabla\cdot}$ scalar. Applying two consecutive operators always gives zero.

Example 3: Test with Curl, Then Find the Potential

Example 3

Determine whether $\vec{F} = \langle y^2z^3,\ 2xyz^3,\ 3xy^2z^2\rangle$ is conservative. If so, find f.

Compute $\nabla \times \vec{F}$. If it's $\vec{0}$, integrate to find f.

Homework

Section 16.5: 1, 3, 5, 7, 9, 11, 15, 17, 19, 21--24, 39